

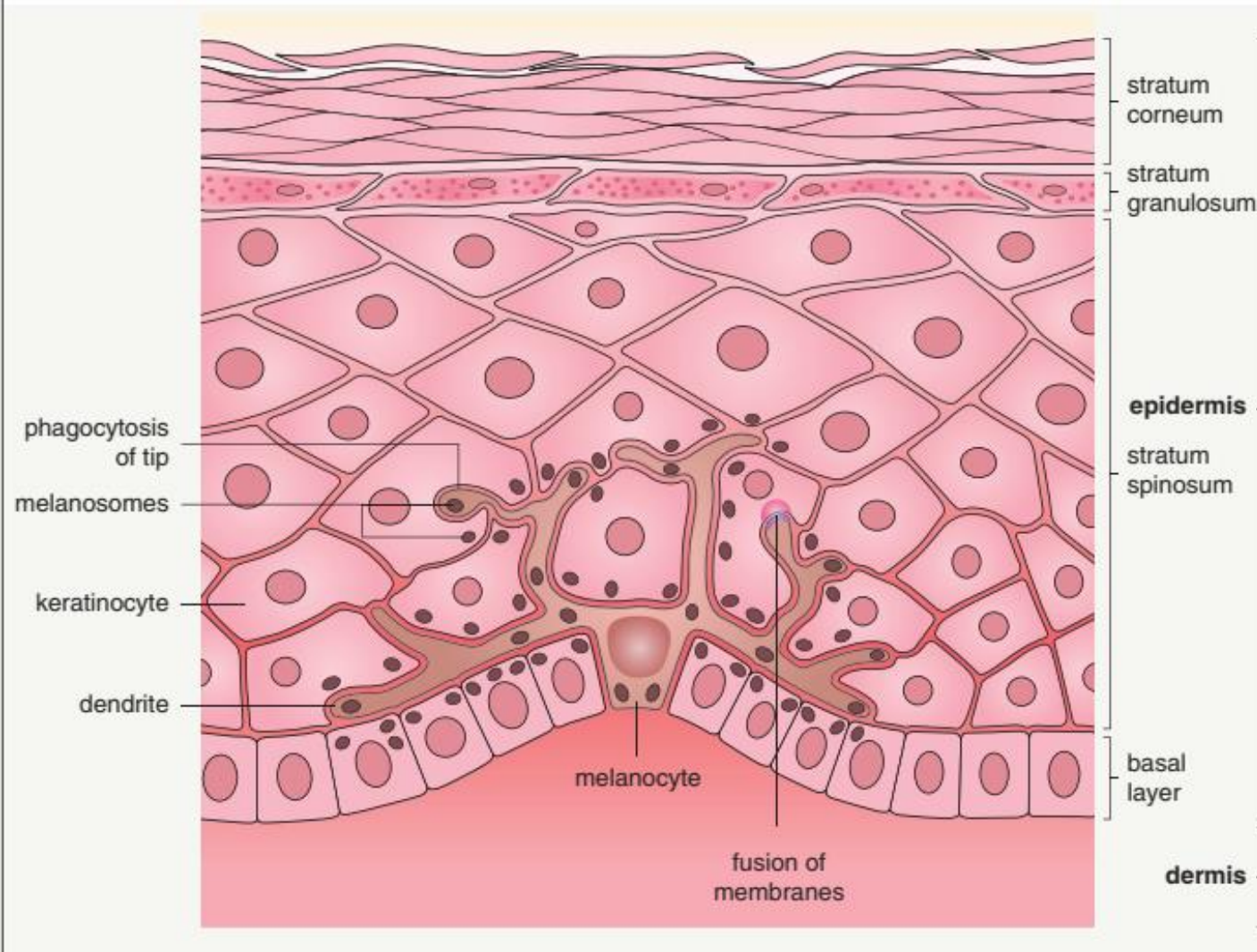
PIGMENTARY DISORDERS:

Dr eman gomaa

Consultant dermatologist

1

A MELANOCYTE RESIDING IN THE BASAL LAYER OF THE EPIDERMIS



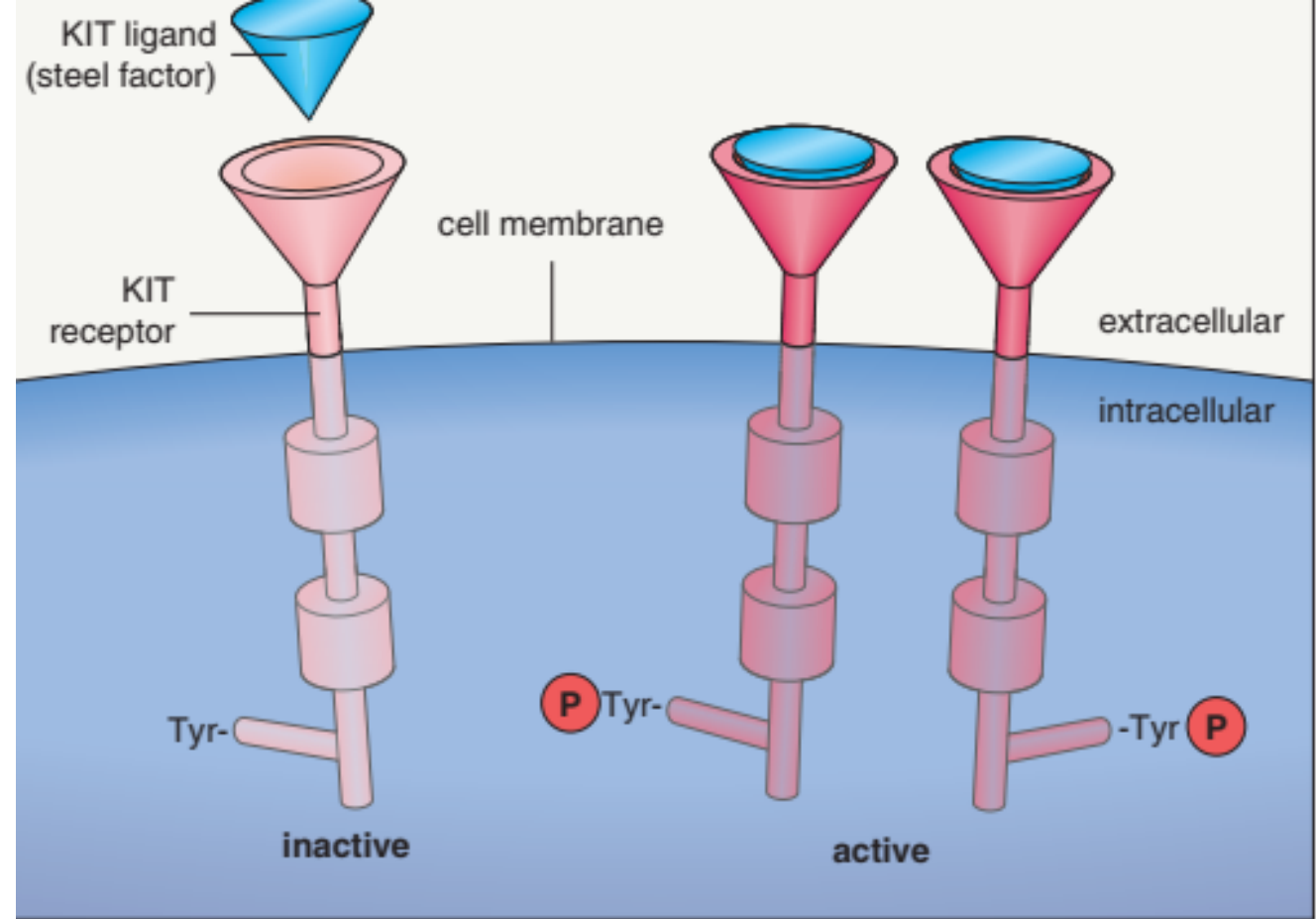
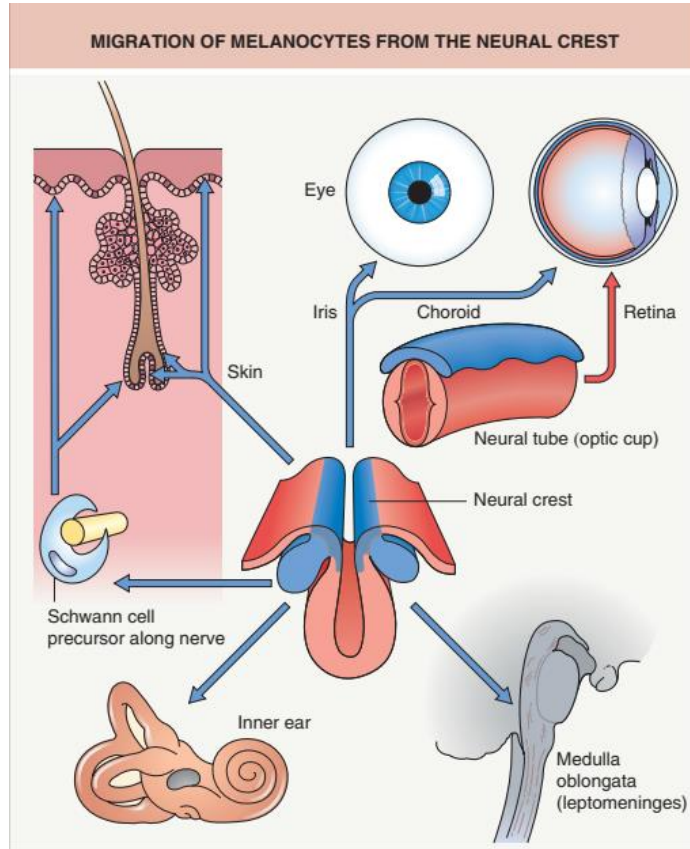


Fig. 65.2 Activation of the KIT receptor on melanocytes. Because the KIT receptor is a tyrosine kinase receptor, it has the ability to phosphorylate the tyrosine residues of other proteins as well as itself, i.e. autophosphorylation. Heterozygous germline mutations in *KIT* that prevent the activation of the KIT receptor by KIT ligand, also referred to as steel factor, lead to piebaldism, whereas somatic activating mutations in *KIT* are seen in patients with mastocytosis as well as melanomas arising in acral sites, mucosae, and chronically photodamaged skin (see Figs 118.2 & 113.1). A form of familial progressive hyperpigmentation with or without hypopigmentation can result from heterozygous gain-of-function germline mutations in the gene that encodes KIT ligand⁴⁹. P, phosphorylation; Tyr, tyrosine.

FORMATION OF A MELANOSOME AND PATHOPHYSIOLOGY OF OCA1

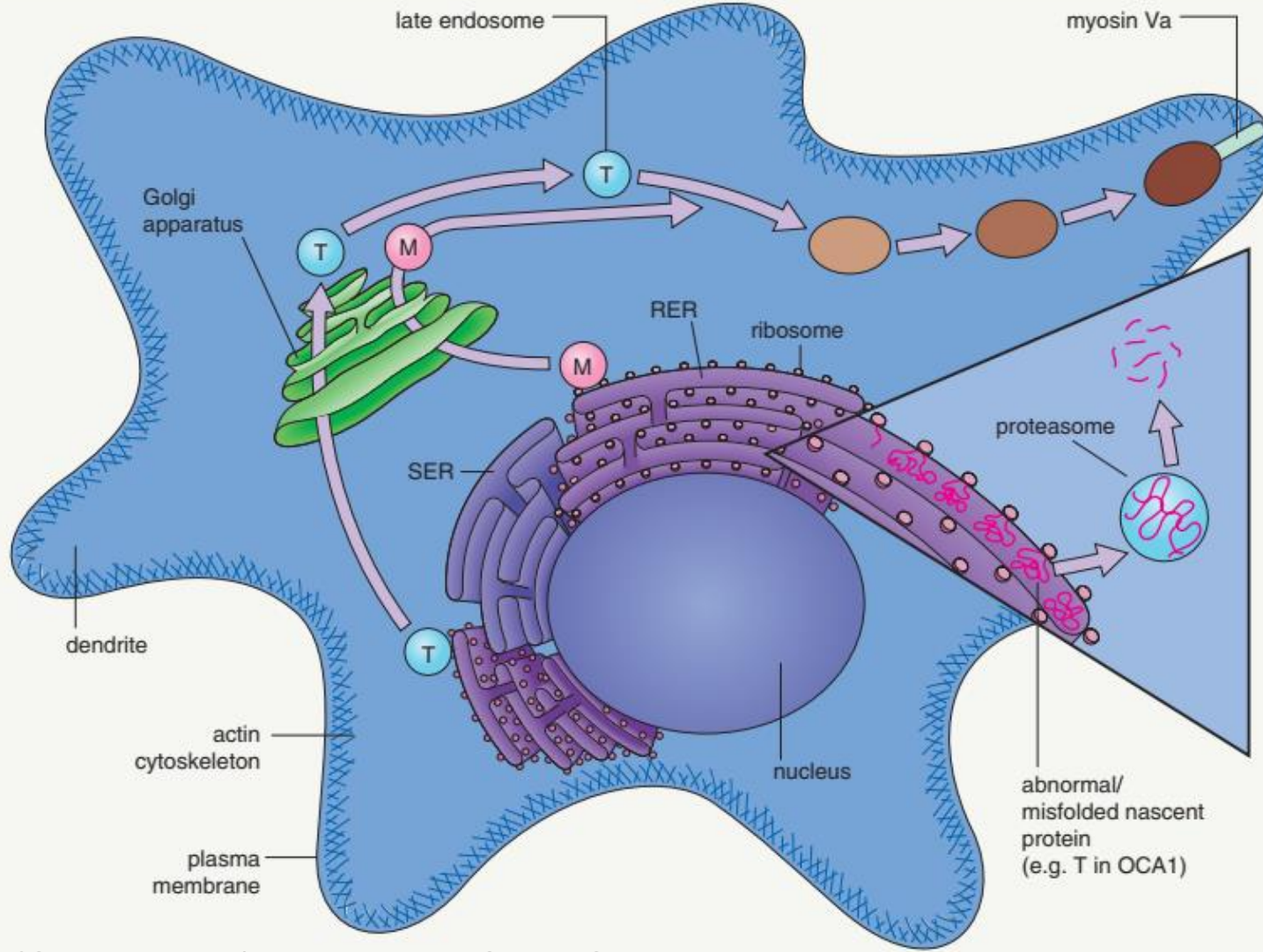
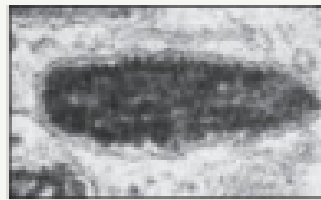
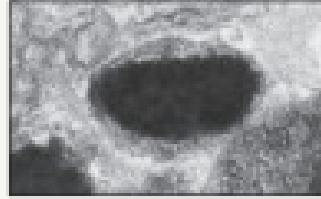


Fig. 65.7 Formation of a melanosome and pathophysiology of oculocutaneous albinism type 1 (OCA1). After glycosylation and processing within the endoplasmic reticulum (ER) and Golgi apparatus, several of the enzymes involved in the formation of melanin (including tyrosinase) are packaged in vesicles and then combine with the matrix proteins (e.g. PMEL/PMEL17/gp100). As more melanin is deposited within the melanosomes, they migrate into the dendrites in preparation for their transfer into neighboring keratinocytes. In patients with OCA1 and OCA3 (triangular insert), tyrosinase polypeptides are retained too long within the lumen of the RER and then become targets for destruction within proteasomes. Thus, they are not incorporated into melanosomes. The primary matrix protein is PMEL/PMEL17/gp100 (recognized by HMB45 antibody); formation of the matrix fibers requires cleavage of this protein by a proprotein convertase⁵¹. Sorting of proteins to the correct organelles is a complex process and it requires regulators such as adaptor protein complex 3 (AP-3; see Fig. 65.8); in the case of tyrosinase, specific dileucine residues in its cytoplasmic tail aid in the sorting. M, matrix proteins;

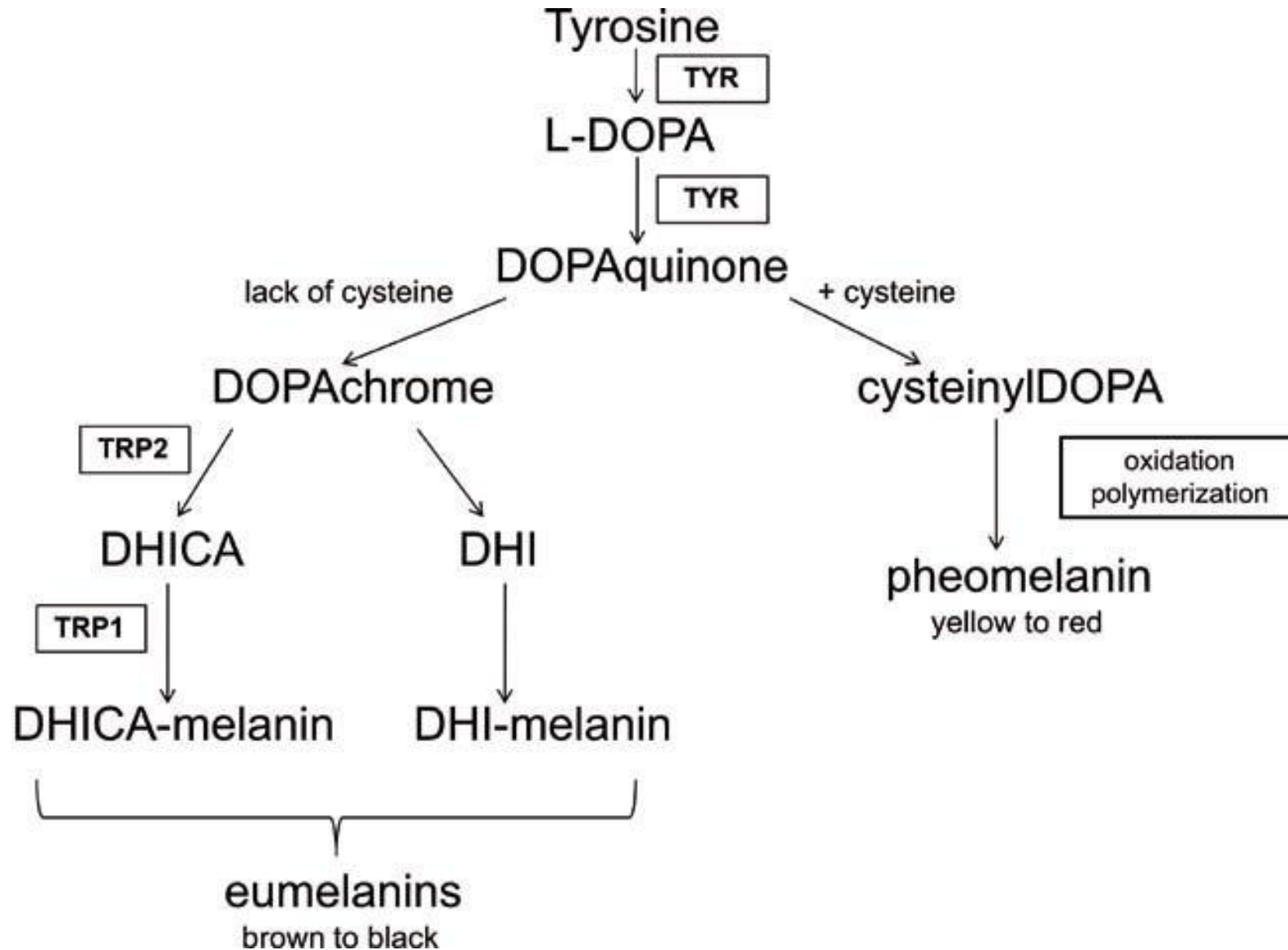
FOUR MAJOR STAGES OF EUMELANIN MELANOSOMES

Stage	Description	Electron micrographs
I	Spherical; no melanin deposition	
II	Oval; obvious matrix in the form of parallel longitudinal filaments; minimal deposition of melanin; high tyrosinase activity	
III	Oval; moderate deposition of melanin; high tyrosinase activity	
IV	Oval; heavy deposition of melanin; electron-opaque; minimum tyrosinase activity	

ATION OF PREDOMINANT MELANOSOMAL STAGES WITH LEVEL OF CUTANEOUS PIGMENTATION

entation of skin	Predominant melanosomal stages	
	Melanocytes	Keratinocytes
	II, III	Occasional III
m	II, III, IV	III, IV
	IV > III	IV

MELANOSOMES IN LIGHTLY PIGMENTED VERSUS D PIGMENTED SKIN		
	Lightly pigmented skin	Darkl skin
Melanization	Stages II, III	Stage
Size (diameter)	0.3–0.5 microns	0.5–0.
Number per cell	<20	>200
Distribution of melanosomes within the lysosomes of keratinocytes	Groups of 2–10 	Single 
Degradation	Fast	Slow



Hypopigmentation

Congenital:

- Oculocutaneous albinism
 - A. Hermansky–Pudlak syndrome
 - B. Chediak–Higashi syndrome
- 2. Piebaldism
 - A. Waardenburg syndrome
- 3. Griscelli Syndrome
- 4. Nevus Depigmentosus
- 5. Hypomelanosis of Ito
- 6. Ash leaf of Tuberous Sclerosis

Acquired

- Vitiligo
- 2. Idiopathic Guttate Hypomelanosis
- 3. Progressive Macular Hypomelanosis
- 4. Halo Nevus
- 5. Melanoma Associated Leucoderma
- 6. Nevus Anemicus
- 7. Wornoff's ring
- 8. Bier Spots
- 9. Post Inflammatory
- 10. Nutritional and Metabolic
- 11. Chemical Agents

VITILIGO

VITILIGO

- Acquired disorder characterized by circumscribed depigmented macules and patches that result from the **loss of functional melanocytes**
- Affects 0.5–2% of the world's population
-

pathogenesis

Genetic
+ve FH in some cases.
Increased in monozygotic
twins

Environmental
Koebner
Stress
Ultraviolet

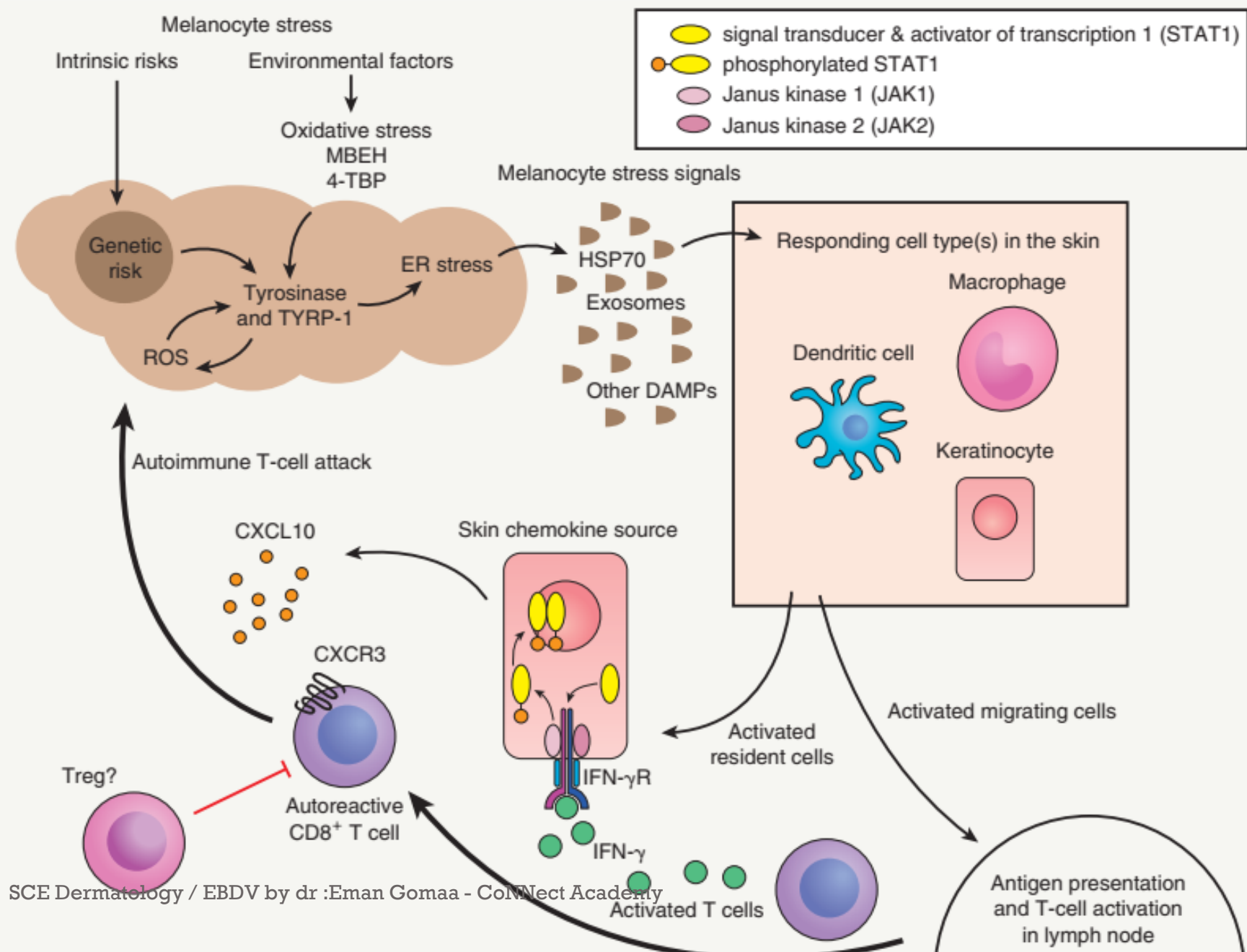
Immunological
Autoantibodies against
melanocyte antigens
T cells against
melanocyte antigens (

Pathogenic hypothesis
**Intrinsic defect in
Melanocytes structure,
function and adhesion**

**Melanocyte self
destructive theory**

**Melanocyte self
destructive theory**

PROPOSED MODEL FOR THE PATHOGENESIS OF VITILIGO



C/P

- totally amelanotic (milky white) macules or patches surrounded by normal skin
- Usually asymptomatic,
- Vitiligo may develop anywhere on the body
- **Koebner phenomenon:**
 - o Development of vitiligo in sites of trauma, such as a surgical excision, burn, or abrasion.







Inflammatory vitiligo. An erythematous inflammatory border is

CLASSIFICATION

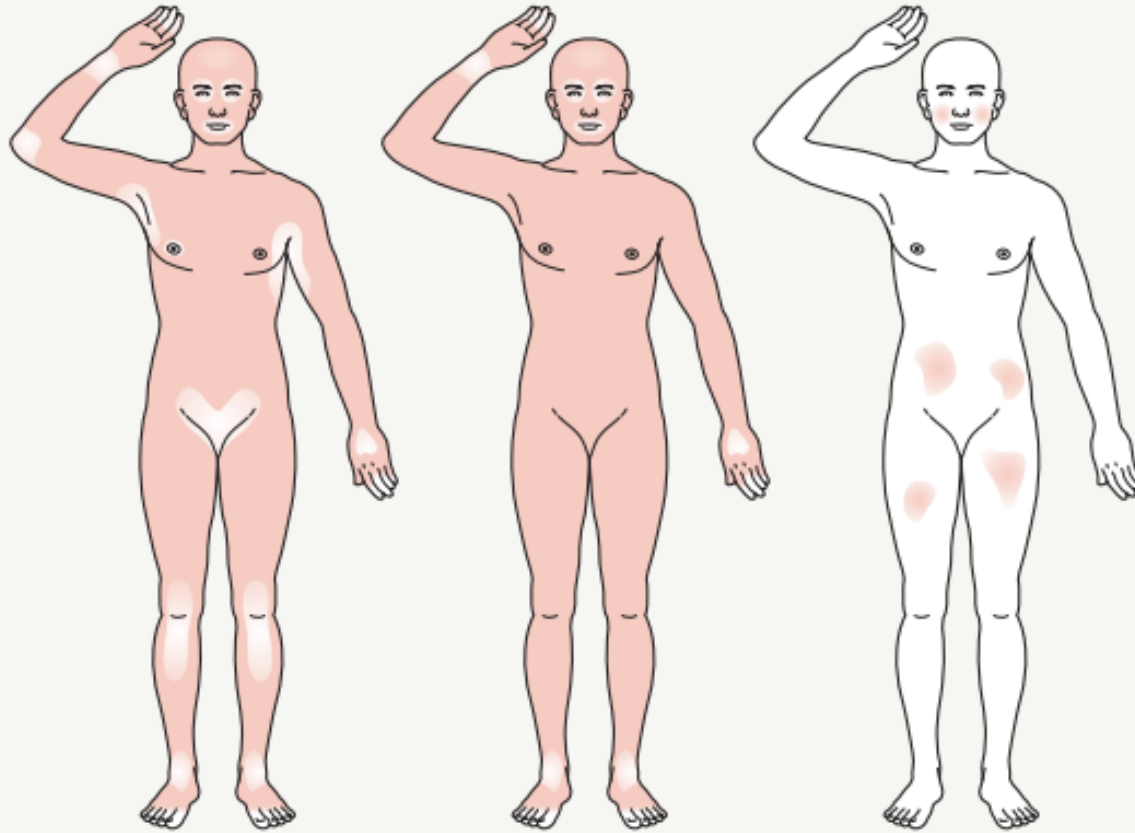
1. Distribution
 - Segmental
 - Non segmental
 - Mixed

- **Segmental vitiligo**
usually reaches its full extent within 1–2 years and remains restricted to the initial segmental area

2- extent
Localized : , , , ,
Generallize :

DISTRIBUTION PATTERN OF AMELANOTIC SKIN LESIONS IN VITILIGO

Generalized Vitiligo

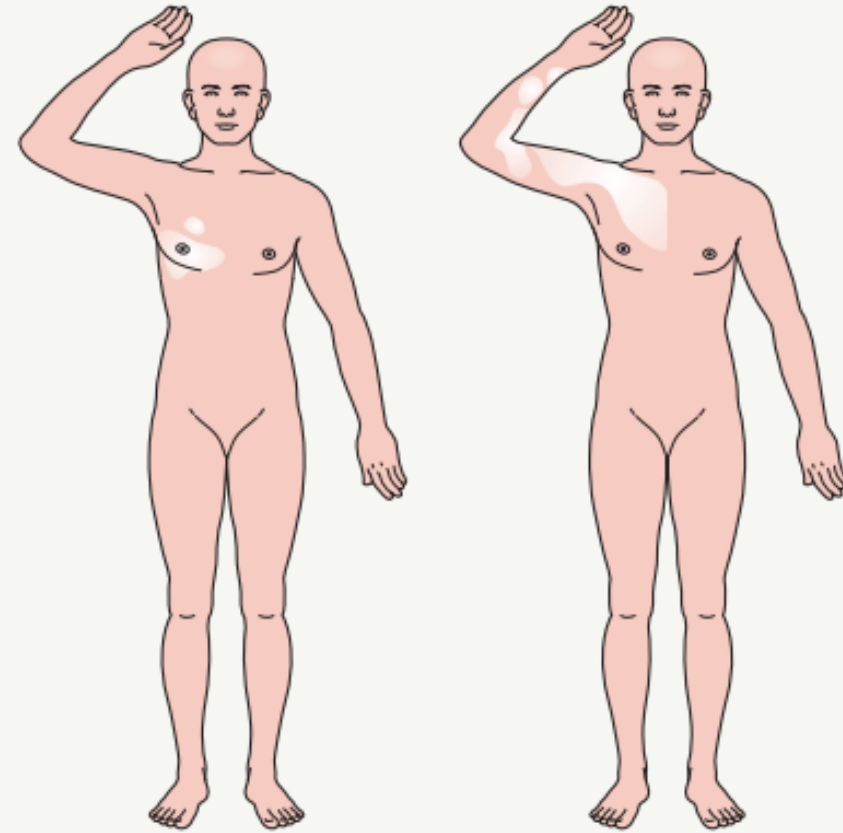


Vulgaris

Acrofacial

Universal

Localized Vitiligo



Focal

Segmental



Fig. 66.5 Segmental vitiligo. Under normal light, vitiligo can be subtle in lightly pigmented individuals. The clue to the diagnosis is the poliosis of the



Fig. 66.7 Segmental vitiligo. Unilateral band of depigmentation on the face, the most common location for segmental vitiligo. Note the pigmented and depigmented hairs within the affected area. *Courtesy, Kalman Watsky, MD.*



SCE Dermatology / EBDV by dr :Eman Gomaa - CoNNeCT Academy
Genital vitiligo. Complete loss of pigment on the penis and



iligo. Depigmentation of the volar wrists as well as the palm

ASSOCIATION :

- Other autoimmune disease:
 - Thyroid , LE
- Ocular :
 - **Vogt–Koyanagi–Harada (VKH) syndrome:**
 - (1) Uveitis
 - (2) Aseptic meningitis
 - (3) Otic involvement (e.g. dysacusia)
 - (4) Vitiligo, especially of the face or sacral region, and associated poliosis
 - **Alezzandrini syndrome:**
 - o Unilateral whitening of scalp hair, eyebrows, and eyelashes as well as ipsilateral depigmentation of facial skin and visual changes

- **Pathology:**
 - In lesions of vitiligo, melanocytes are typically absent or present in very small numbers

- The epidermal melanocyte density can be assessed with:
 - **Melan-A (MART-1):** melanocyte-specific immunohistochemical stains
 - **DOPA:** which detects tyrosinase activity



(a)



(b)

o on

- The aims are
- stabilization of the depigmentation process and
- Repigmentation
- - The face, neck, mid extremities, and trunk tend to have the best response to therapy
- - While the distal extremities and lips are the most resistant to treatment

TREATMENT

topical

Systemic

Phototherapy

Surgical Therapy

laser

DEPIGMENTATION

antioxidant

Topical corticosteroids

Useful for localized areas of vitiligo

Topical Calcineurin Inhibitors



Cream 30 g Ointment 30 g Scalp Application 25 mL	A very potent topical treatment for difficult-to-manage steroid-responsive dermatoses ³⁻⁵	 Dermovate® Clobetasol propionate
Cream 30 g Ointment 30 g Scalp Application 100 ml	A potent topical treatment with tried and trusted efficacy in steroid-responsive dermatoses ⁶⁻⁸	 Betnovate® Betamethasone valerate
Cream 15 g Ointment 15 g	A potent topical treatment designed for efficacy and tolerability in steroid-responsive dermatoses ⁹⁻¹³	 Cutivate® Fluticasone propionate
Cream 30 g	A moderately potent topical treatment more effective than hydrocortisone 1.0% in steroid-responsive dermatoses ¹⁴⁻¹⁸	 Eumovate® Clobetasone butyrate



Systemic Steroids

- Oral minipulse is used to arrest disease progression
- Dexamethasone tablets are given 2 consecutive days per week (weekend therapy)
- 4 months the full dose followed by tapering over 2 months



Phototherapy
NB-UVB

- **The first-line treatment for adults and children ≥ 6 years of age with generalized vitiligo**



Excimer Laser And Lamp

- wavelength of the 308 nm excimer laser and lamp is close to that of NB-UVB
- Localized patches of vitiligo are treated one to three times weekly with the excimer laser, typically for a total of 24 to 48 sessions
 - Not useful in large areas



Surgical

1. Patients who fail to respond to medical therapy
2. **Stable disease for ≥ 6 months**
3. Absence of the koebner phenomenon
4. No tendency for scar or keloid formation

- 1-Minigrafting:
- 2. Suction blister epidermal grafting
- 3. Grafting of cultured autologous melanocyte



Antioxidants

Ginko Biloba

Combination Therapy

Psychological support

Depigmentation

20% monobenzyl ether of
hydroquinone (MBEH),
applied once to twice daily



Idiopathic Guttate Hypomelanosis



Progressive Macular Hypomelanosis



Nevus Anemicus

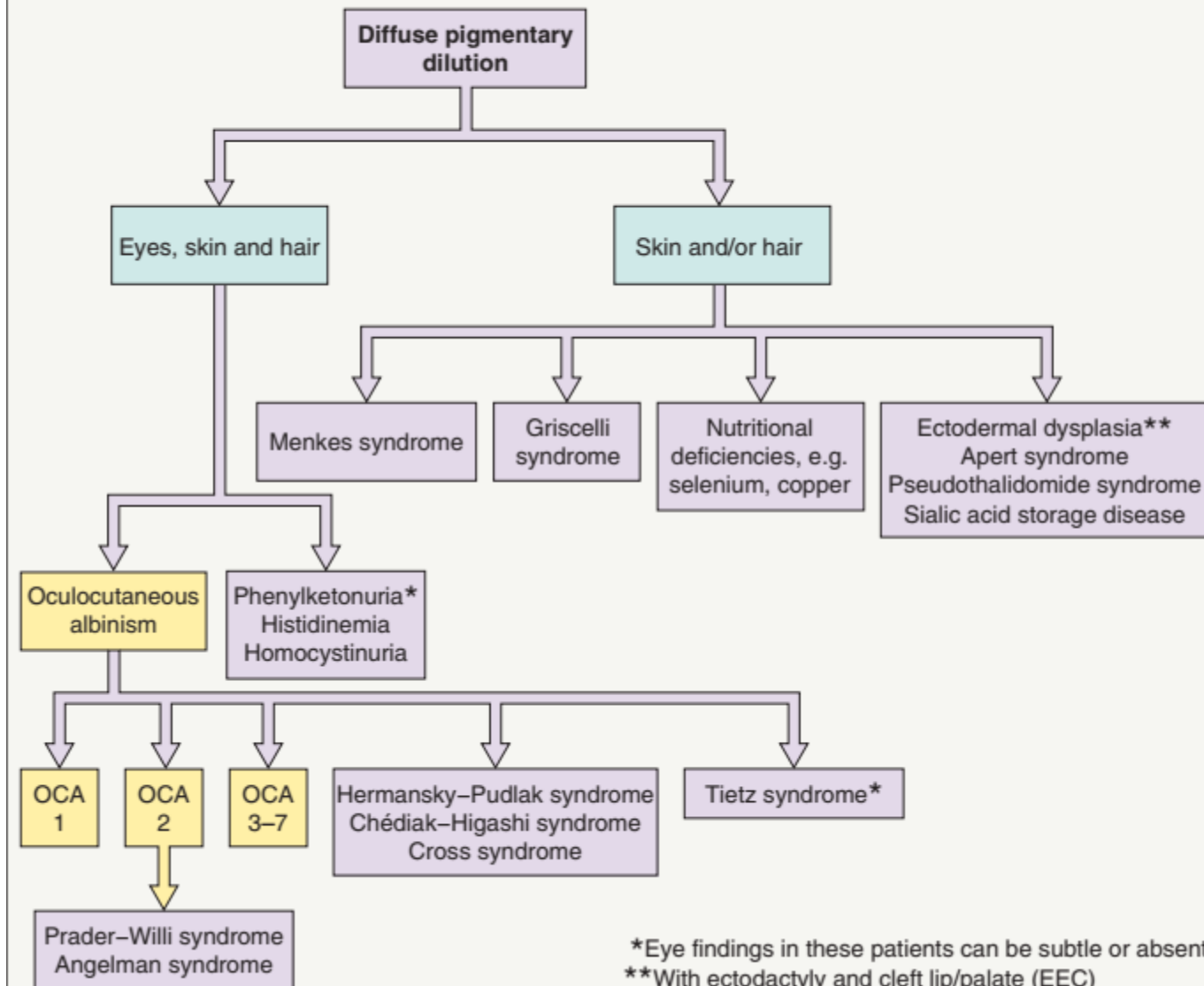


Wornoff's ring

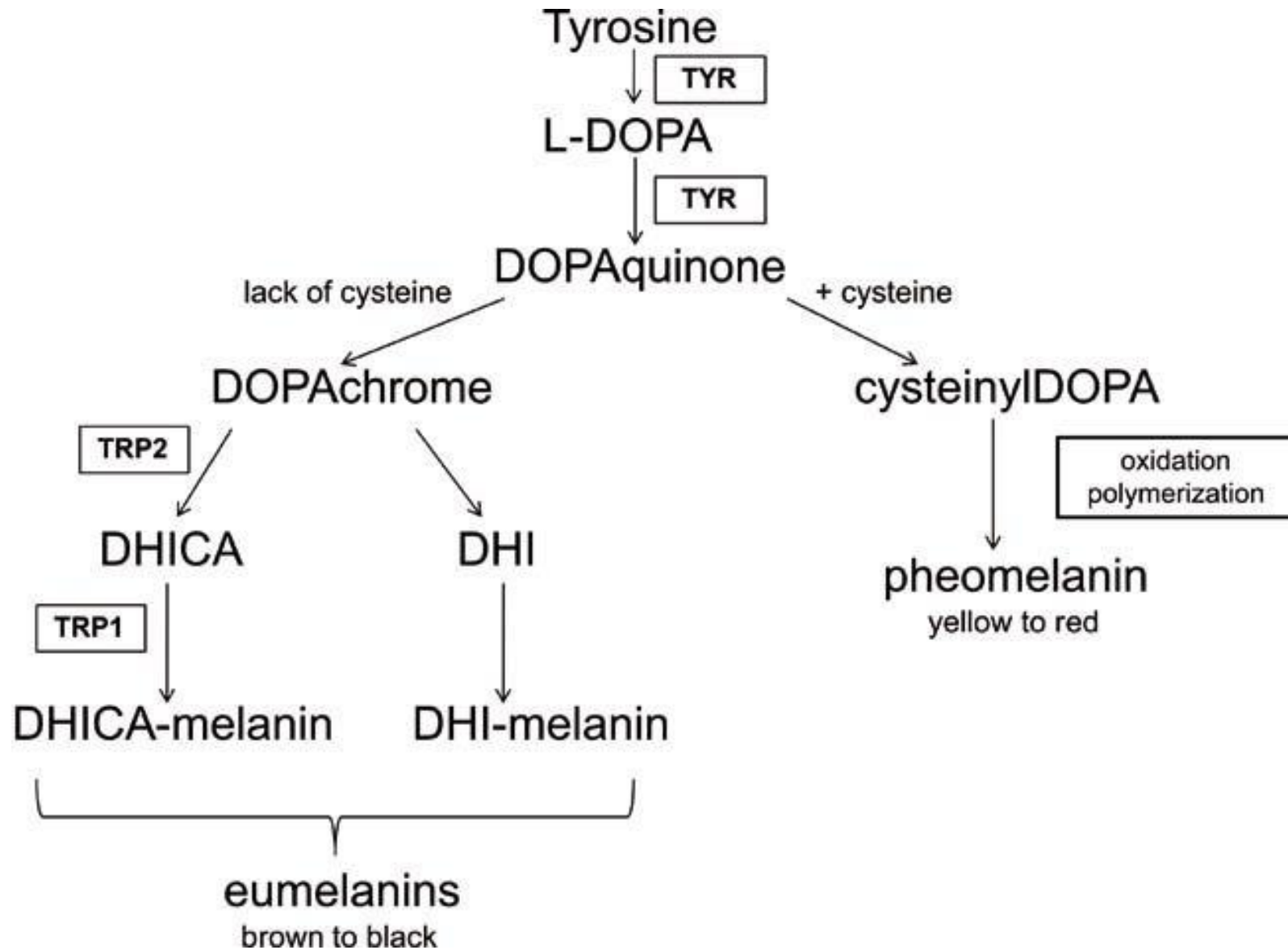


Bier Spots

DIFFERENTIAL DIAGNOSIS OF DIFFUSE PIGMENTARY DILUTION WITH AN ONSET AT BIRTH OR DURING INFANCY



CONGINITAL , HEREDITRY HYPOMELANOSIS



OCULOCUTANEOUS ALBINISM

A GROUP OF GENETIC DISORDERS CHARACTERIZED BY DIFFUSE PIGMENTARY DILUTION DUE TO A PARTIAL OR TOTAL

ABSENCE OF MELANIN PIGMENT WITHIN MELANOCYTES OF THE SKIN, HAIR FOLLICLES, AND EYES

OCA1	OCA2	OCA3	OCA4	OCULAR ALBINO
TYROSINASE deficiency 2 types : A :COMPLETE ABSENCE - No melanin - White hair , gray eye	Defect in P-PROTEIN	TYRP1	SOLUBLE CARRIER SLC15 A2	XLR – BOYS ONLY OCULAR
type 2 :reduced tyrosinase				

OCA2A



Fig. 66.10 Oculocutaneous albinism type 2 (OCA2). African patient with obvious squamous cell carcinomas on the cheek as well as multiple pigmented

WAARDENBURG SYNDROME

Waardenburg Syndrome

Waardenburg syndrome (WS) is a rare autosomal dominant or less often autosomal recessive disorder that is characterized by various combinations of the following features⁹⁷:

- achromia of the hair, skin, or both in the same pattern as piebaldism (see above)
- congenital deafness
- partial or total heterochromia irides (including isohypochromia)
- medial eyebrow hyperplasia (synophrys)
- a broad nasal root
- dystopia canthorum (increased distance between the inner canthi, with a normal interpupillary distance).



PIEBALDISM

- AD
 - Mutations in the *KIT* proto-oncogene (required for the normal development of melanocytes)
 - Lesions are present at **birth**, remain **stable**

1. White forelock (90%)

○ Midline in location, triangular or diamond-shaped

2. Leukoderma

○ Irregular in shape, well-circumscribed, and milk-white in color

○ Favors the central anterior trunk, mid extremities, central forehead

○ Contain hyperpigmented macules or patches

- **TTT:** autografting of melanocytes, Camouflage



Fig. 66.13 Piebaldism. White forelock and characteristic triangular amelanotic

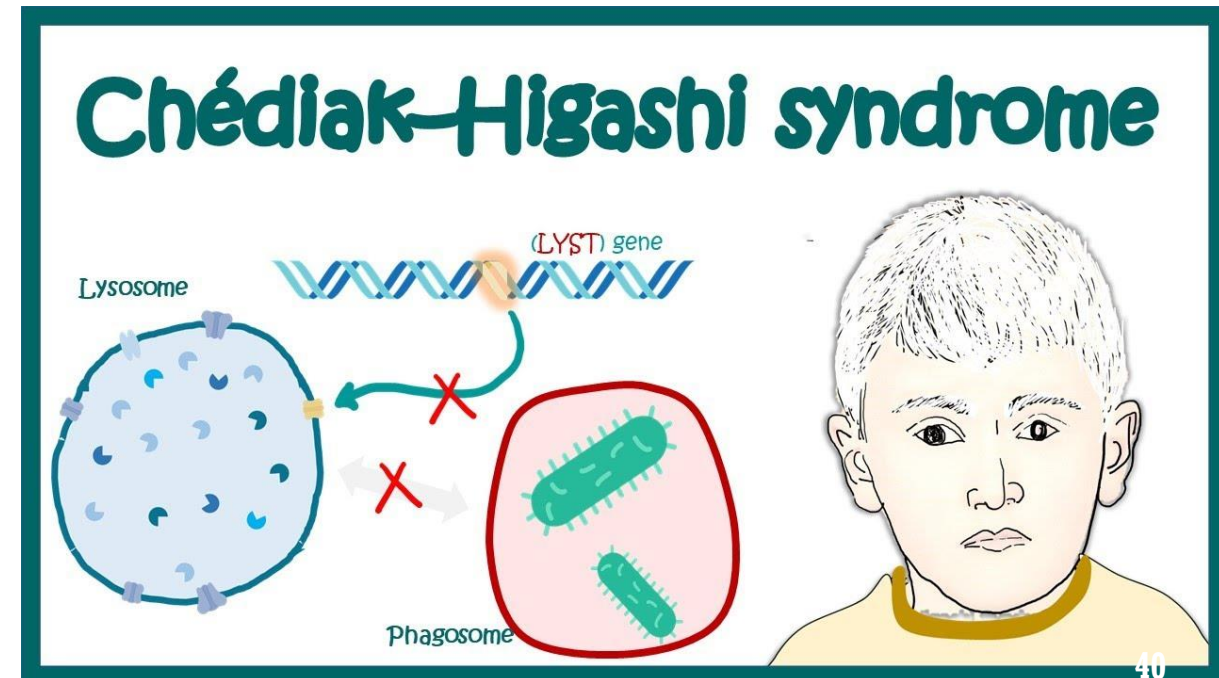
CHEDIAK–HIGASHI SYNDROME

■ Pathogenesis:

- Block of transfer of melanosome due to formation of macromelanosomes
(mutations in the *lysosomal trafficking regulator* gene)

Clinical Picture:

1. Oculocutaneous **albinism** with photosensitivity
2. Hair: Silver grey
- 3. Ocular** defects
4. Susceptibility to **infections**



GRISCELLI SYNDROME

Due to mutations in **genes responsible for melanosome transfer** from melanocytes to keratinocytes

Clinical Picture:

1. **Pigmentary dilution** of the skin and silvery-gray hair due to pigment clumping within melanocytes
2. **Neurologic** impairment (with type 1)
3. **Immune** abnormalities (with type 2)





Nevus depigmentosus



Hypomelanosis of Ito

